SCI-MAP

TolTEC Ordinary Mapmaking

and Observation Coaddition

Scientific Basis, Response, and Validity

Scientific contract version: v0.1

Science-team rationale revision: r0.6

Status: integrated scientific authority draft

2026-08-27

This document explains the map estimator, its response and uncertainty, the meaning of support and validity, and the scope of observation coaddition for TolTEC scientists. The separate formal contract remains the normative authority for the 52 requirements, 25 predictions, exact states, provenance, and conformance procedure.

Scientific owner: Grant Wilson

Executive summary and current status

For a map pixel p, SCI-MAP forms the coefficient-weighted average

$$\widehat{m}_p = \frac{\sum_i G_{pi}\gamma_i z_i}{\sum_i G_{pi}\gamma_i} = \frac{N_p}{Q_p}, \tag{1}$$

using only realized PTC samples that are available under PTC's declared identity and scope, have valid ALIGN/AST coordinates, and pass MAP's admission conditions. Here G_{pi} describes how sample i is placed in its unique containing pixel, and γ_i is the exact PTC-owner-selected MAP-facing coefficient. No such family is yet selected, so r0.6 defines the route but withholds ordinary numerical MAP production until that upstream gate closes. Pixels without adequate valid support are unavailable; they are not measurements of zero sky brightness.

For compatible observation maps on one common grid, the v0.1 coadd is

$$\widehat{s}_p = \frac{\sum_o u_{op}\widehat{m}_{op}}{\sum_o u_{op}}, \tag{2}$$

where the named v0.1 uniform family sets dimensionless $u_{op} = 1$ for every admitted observation row. This is equal-observation averaging, not precision. An observation bundle is admitted atomically: its signal, coordinates, response, uncertainty, validity, and provenance must be mutually compatible before any of its valid pixels contribute. Pixel-level support then determines where the admitted observation contributes.

Both equations are linear only *conditional on fixed* sample and detector membership, projection, coefficients, support policy, response state, geometry/WCS, calibration, and upstream processing. If any of those states is estimated from the same data, unconditional bias or covariance requires a joint model or resampling result.

Intended scientific estimator	Positive-coefficient ordinary map and compatible common-grid observation coadd, Equations (1)–(2).
Upstream input status	MAP always consumes a realized positive-rank PTC r0.5 product; there is no CAL fallback, PTC-disabled MAP route, or inferred no-op PTC product. The exact PTC coefficient family and admitted numerical `coverage_cut` domain remain hard blockers, so no ordinary numerical route is generally authorized by this draft.
Implementation conformance	Not assessed in this authorship revision.
Response/representation validation	Not established; response injections and WCS round trips remain required evidence.
Observational validation	Not established; amplitude, shape, position, covariance, and repeatability require astronomical controls.
Production readiness	Not claimed and cannot be inferred from the algebraic contract.

Route level	Status	Principal remaining gate
Conditional estimator	Defined	Equations and scientific meanings only.
Numerical observation map	Unavailable	PTC coefficient family and admitted numerical `coverage_cut` state.
Response-bearing map	Product-dependent	Exact sufficient fixed-state, PTC procedure, or full-chain response.
Uncertainty-qualified map	Product-dependent	Exact required covariance representation and terms.
Compatible-grid coadd	Conditional	Source-closed admitted observation maps and exact coadd profile.
Reprojection/mosaic	Not authorized	Separate owner decision and contract.

Contents

1 What SCI-MAP estimates

SCI-MAP estimates a map-domain average of the *realized PTC-transformed* calibrated-x-derived nonpolarimetric total-intensity-equivalent quantity $z_i = Z_i^{\mathrm{PTC}}$. Neither `mJy/beam` nor a STOKES token promotes it to Stokes I. It does not estimate unconvolved sky truth. The physical meaning of the result is therefore inherited from the complete upstream CAL-to-PTC chain: what the ordinary `xs` sample represents, how it was calibrated and transformed, what sky coordinate it has, and whether the required PTC product exists under its declared producer-local validity.

The numerical signal scale is the declared top-of-atmosphere, point-source-equivalent mJy beam^{-1} convention inherited from SCI-CAL with the exact originating fixed-nominal-beam/template identity. MAP does not create absolute calibration, choose a beam convention, or complete an unresolved broadband photometric definition. A numerical map may consequently exist while literal point-source-peak interpretation remains unavailable because the response for the declared processing chain is incomplete. The same caution applies to extended-source fidelity, integrated flux, and absolute offsets.

For one exact frozen upstream and MAP state, the linear response is the fixed-state sky-to-PTC operator followed once by the map operator. Writing the selected processed-sample vector as $\boldsymbol{z} = (z_i)$,

$$\widehat{\boldsymbol{m}} = A_{\mathrm{MAP},\Pi}\boldsymbol{z}, \qquad R_{\mathrm{fixed},\Theta} = A_{\mathrm{MAP},\Pi}H_{\mathrm{fixed},\Theta}, \tag{3}$$

where $H_{\mathrm{fixed},\Theta}$ binds source domain, processed-sample codomain, units, parent, membership, support, coefficient state, and response basis. Preserving a constant *sample vector* does not prove preservation of a constant sky mode: an upstream baseline, common-mode removal, or filter may already have removed it.

A stored kernel is a fixed-state realized response only when its template, sample-domain parent, and processing lineage prove Equation (3). Otherwise it is a tracer or has an explicit limited or unavailable status. That state does not invalidate the numerical map or prohibit later analysis; it only limits the claims made by the original bundle. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity.

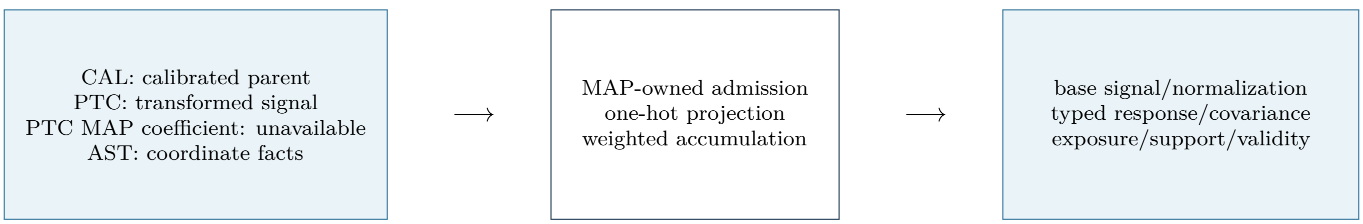

Figure 1: SCI-MAP transforms declared upstream quantities; it does not repair or reinterpret them.

2 From samples to pixels

An occurrence i is not merely time and detector. It binds the observation, detector occurrence and UID, stable RTC output sample n, exact PTC product/application generation, segment, and array/network/group. Time is an attribute; input-column, row, and container positions are locators. The AST coordinate is exactly `SCI-AST:rtc_output_grid_coordinates@1` for the same n and parent chain. MAP never joins signal and coordinate by time, row, cardinality, or numerical coordinate equality.

MAP owns `SCI-MAP:one_hot_containing_pixel@1`. A finite coordinate inside the target extent belongs to the unique pixel whose cell contains it under half-open lower-inclusive, upper-exclusive bounds. There $G_{pi} = 1$; it is zero for every other pixel. An internal upper edge belongs to the adjacent cell, while the outer upper boundary and every out-of-grid coordinate contribute nowhere. Fractional projection is deferred until a complete kernel, support, normalization, conservation, boundary, response, and covariance family is separately authorized. AST may materialize G, but the artifact must bind both the AST coordinate parent and the exact MAP plan.

PTC owns the MAP-facing coefficient γ_i. Frozen PTC r0.5 defines the required coefficient declaration but leaves selection open in PTC-OD-010. MAP therefore does not infer unity, loading, sensitivity, scatter, or inverse variance: no ordinary numerical MAP route exists until the PTC/scientific owner selects and versions one exact family. A later coefficient generation does not modify an earlier transformed product, and cross-generation use needs an explicit compatibility decision.

The representation-independent handoff is `SCI-PTC_TO_SCI-MAPv0.1/r0.1`. MAP owns the occurrence-level policy `SCI-MAP:map_upstream_admission@2`; the continuing VAL Registry binds it and VAL Core evaluates it. The policy requires the exact PTC output-retention permission and preserved ancestry, excludes direct synthesized/replaced representatives and unresolved required facts for this use, carries inherited influence without turning it into a universal veto, and records response and uncertainty roles. Pixel placement and final contribution remain MAP estimator actions.

For a future compatible coefficient family the effective contribution is $G_{pi}\gamma_i$. Membership is the conjunction of ten separately typed Boolean pass predicates derived from ten retained typed gate records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC disposition; MAP upstream admission; AST coordinate validity; finite transformed signal; finite positive coefficient; admitted projection/boundary; and MAP-local contribution plus required companions. Unknown, conflict, ineligible, unavailable, invalid, and not-requested all project to estimator nonmembership where applicable while their full T/F/U/C or VAL request/applicability/eligibility/realization record, causes, provenance, and failure scope remain intact. The payload is never screened by multiplication with a numeric zero mask. MAP then accumulates

$$N_p = \sum_{i \in \mathcal{C}_p} G_{pi}\gamma_i z_i, \qquad Q_p = \sum_{i \in \mathcal{C}_p} G_{pi}\gamma_i. \tag{4}$$

The numerator is not yet a sky estimate, and the denominator is a gridding normalization by default. Division occurs only where the adopted support policy permits a scientific map value.

3 Projection, response, and removed modes

The one-hot family closes the initial G_{pi} route without making a fractional-kernel claim. It also keeps geometry distinct from eligibility: AST-geometric incidence can exist when the PTC signal is unavailable; MAP-route-candidate incidence additionally requires the realized route and upstream admission; estimator contribution adds every remaining numerical and MAP-local gate.

Fixed-state, PTC full-procedure, and full-chain response are different scientific objects. The fixed-state family uses Equation (3) with every upstream and MAP choice frozen. PTC full-procedure response instead compares two complete PTC runs,

$$\Delta \boldsymbol{z}_{\mathrm{FP}}(\delta \boldsymbol{s}; \alpha) = \mathcal{P}_{\mathrm{PTC}}(\boldsymbol{s} + \delta \boldsymbol{s}; \alpha) - \mathcal{P}_{\mathrm{PTC}}(\boldsymbol{s}; \alpha),$$

and separately records changes in masks, groups, rank, loading subspace, classifications, coefficient state, stop cause, and support. The parameter α binds amplitude, perturbation side, source state, location, and domain. If MAP membership, coefficient, projection, and support stay fixed, MAP may apply $A_{\mathrm{MAP},\Pi}$ to that finite difference. If MAP is re-resolved, the result is a separately named full-chain procedure response, not one fixed H matrix.

A realized PTC-grid companion starts at MAP and receives the fixed MAP operator exactly once. If any signal member lacks a required exact companion, MAP keeps the complete signal membership and reports response limited or unavailable; it does not construct a smaller hidden subset.

The PTC nonrestored additive reference λ, fitted correlated removed component $\widehat{U}_{\text{corr}}$, total removed component $U_{\text{total},\Theta}$, realized removed-subspace identity, fixed-state null space, and any full-procedure invariant/unidentifiable modes are separate facts. They are not automatically residual bias. The fixed-state perturbation statement is Equation (3); r0.6 makes no unnamed μ_0 or residual conditional-bias claim.

4 Weights and uncertainty

For fixed upstream and map state, uncertainty propagates through the same map operator as the signal:

$$C_m = ANA^{\mathsf{T}}, \tag{5}$$

where N is the declared PTC-domain processed-sample noise covariance when it is available for the asserted domain. Equation (5) retains cross-pixel terms created by shared samples and any correlations already present in the timestream.

Four quantities answer different questions:

1. Q: the gridding denominator that normalizes the map;
2. a formal marginal inverse variance: valid only under its stated coefficient and covariance assumptions;
3. the full conditional map covariance or precision, including correlated pixels when available; and
4. empirical covariance, weights, or statistical significance calibrated by NOI from admitted realizations or data.

The base/unfiltered MAP bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, recoverable from lineage, summarized, incomplete, or unavailable. Missing correlations and nuisance terms are not zeros. Normalization, hits, exposure, and diagnostic weight are not covariance. Absence of a complete covariance model neither invalidates the map nor prohibits later analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity.

The distinction is especially important when sample membership, weights, projection, response, or support is estimated from the same observation. The linear equations remain useful conditional descriptions, but they do not include selection uncertainty. A joint model, resampling study, or other separately justified treatment is needed for an unconditional statement.

Quantity	Owner/status	Science interpretation
Analysis coefficient γ	PTC, family not yet selected	Required gridding coefficient; no numerical MAP until selected; never precision by implication.
Normalization Q	MAP	Denominator of the ordinary estimator; not automatically uncertainty.
Conditional covariance	PTC input, MAP propagation	Fixed-state error covariance through projection and normalization.
Empirical weight/significance	NOI	Requires declared realizations, calibration, and validity evidence.
Calibration/response nuisance	SCI-CAL and response owner	Retained as quantified, lineage-resolvable, or unavailable; never silently zero.

Table 1: Ownership prevents a convenient numerical product from acquiring a stronger statistical meaning.

5 Coverage, support, and validity

The formal contract keeps the following facts separate. For science users they answer the following questions:

Science question	Product	Interpretation
Did a valid AST coordinate reach this pixel?	AST-geometric incidence	Geometry independent of PTC signal availability.
Did an occurrence have the realized PTC route and pass upstream admission?	MAP-route-candidate incidence	Route/use policy before local numerical gates.
How many admitted weighted occurrences entered the estimator?	Estimator contribution	Exact contribution count.
How many complete observations entered this coadd pixel?	Contributing observation count	Observation bundles, not samples.
What detector-time was available before estimator-specific rejection?	Upstream-eligible exposure	Eligible detector-time accounting.
How much detector-time was retained by the estimator?	Retained exposure	Retained accounting, not precision.
Was the denominator large enough to normalize?	Numerical support	Lower support threshold.
Is the pixel inside the adopted science region?	Science-policy support	Stricter policy threshold.
Is the base MAP value scientifically valid under MAP's own contract?	MAP-local base-product validity	Required conjunction of identity, support, finite value, and no MAP-product failure; not downstream eligibility.

Hits are not validity, exposure is not precision, and positive normalization is not sufficient for final validity. A rectangular FITS image may contain storage values outside the scientific domain; the explicit support and validity products determine which pixels are measurements. Unsupported pixels are unavailable, not zero sky.

Exposure begins with ALIGN-owned e_{acq} and e_{vo}, measured in seconds on stable original occurrences. RTC, CAL, and PTC preserve those facts, lineage, and causes. MAP deduplicates by observation, detector occurrence/UID, native original occurrence, and ALIGN mapping generation. It places each positive-e_{vo} original exactly once using that original's own authorized AST coordinate and the one-hot half-open rule.

This is deliberately different from causal influence. One original may influence several RTC/PTC outputs at different map pixels; the influence graph is retained, but its edges neither relocate nor duplicate physical seconds. Upstream-eligible exposure uses originals ancestral to route candidates; retained exposure restricts the original population to ancestors of admitted MAP outputs. Under one-hot placement, one original occupies at most one pixel, or none after boundary loss. Coadded exposure unions observation-scoped original identities after atomic admission. Replacement, synthesis, donor reuse, overlapping filters, decimation, duration, cadence, sample count, hits, Q, γ, and formal precision cannot create or reconstruct exposure. Missing original lineage or own-coordinate authority makes the exposure role unavailable.

> **Provisional two-threshold policy.** A lower threshold permits numerically safe normalization; a stricter threshold defines the adopted science region. Both are scaled from an order statistic of the finite positive normalizations in a declared map population. This is adopted policy, not a law of mapmaking or a proof of completeness, response, or noise quality.

Because the reference normalization is computed from a declared population, changing the map extent or population can change the science mask even when the local samples near a particular pixel are unchanged. That global behavior must be justified and validated for the intended use.

The formal equations compare Q_p with $Q_\star c/10$ and $Q_\star c$. The scientific owner has therefore fixed $c =$ `coverage_cut` as a dimensionless support-policy scalar. SCI-MAP-CI-001 records the resolved dimensional correction. This disposition does not determine the allowed numerical domain or the behavior for negative, zero, non-finite, or greater-than-one values; those questions remain open in SCI-MAP-OD-007.

6 Observation coaddition

V0.1 combines observation maps only when they already represent the same responded quantity on the same pixel grid. Each observation contributes as a complete bundle: signal, normalization, response state, covariance status, exposure, support, validity, WCS, and provenance are checked together.

The permitted placement is a centered integer embedding. A smaller map may sit inside a larger common grid only when each dimension differs by a nonnegative even number of pixels; half the difference is the integer offset. The shifted reference pixel must identify the same world coordinate. An odd shape difference is incompatible and the complete observation is rejected. There is no fractional shift, interpolation, reprojection, or implicit recentering in this ordinary branch.

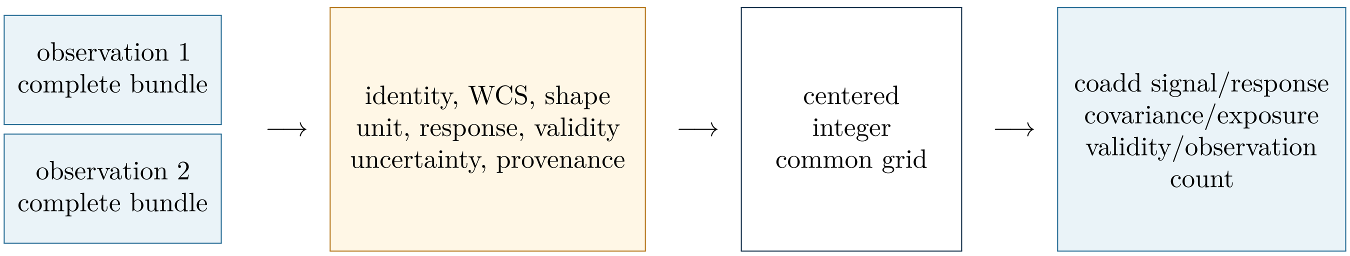

Figure 2: Complete-bundle admission and centered-integer placement. Any failed compatibility check rejects the observation bundle from the coadd before any of its valid pixels contribute.

This restriction is the declared scope of the ordinary v0.1 coadd, not a general scientific argument against mosaicking. Whether observation maps are cropped or padded upstream to a canonical grid, and which future package owns reprojection or mosaicking, are not fixed by the admitted authority and are recorded as SCI-MAP-OD-009. Until then, SCI-MAP performs neither operation.

The coadd-admission profile is `SCI-MAP:observation_coadd_admission@1`. It is a MAP-authored, VAL-governed aggregate profile with separate request, applicability, eligibility, and realization axes. It requires compatible quantity and nominal-beam convention, PTC route/product role, response class, complete AST WCS/grid, support policy, coefficient family, covariance disclosure, null/additive-reference state, required companions, lifecycle, and parentage. The coefficient family is `SCI-MAP:uniform_observation_coadd_coefficient@1`: $u_{op} = 1$, dimensionless, on every admitted observation row. It is an equal-observation coefficient and has no inverse-variance interpretation.

Writing $\widehat{\boldsymbol{m}}_{\mathrm{obs}}$ for the stacked signal on exact observation-row identities, B_{out} is the centered-integer averaging operator. T_{obs} stacks compatible fixed-state member response objects on the same domains, and C_{obs} contains every available within- and cross-observation covariance block. The response-independent base-coadd role makes response advisory and unavailable-compatible. If any member response is unavailable or response-basis-incompatible, the numerical signal membership may remain unchanged, but coadd response is typed unavailable; no smaller hidden membership or zero response is formed. A response-bearing role instead requires every response and an exact compatible source domain, basis, class, unit, normalization, parent, and row identity.

The equation $C_{c,\mathrm{out}} = B_{\mathrm{out}} C_{\mathrm{obs}} B_{\mathrm{out}}^{\mathsf{T}}$ is a complete numerical covariance statement only when every block required by the exact covariance-qualified role is available. Otherwise the coadd publishes an explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable covariance state. Unknown blocks remain unknown, not zero or independence. The denominator is normalization, not precision. Correlated generalized least squares, which may need negative coefficients and a full cross-observation covariance, belongs outside this positive-coefficient branch.

7 Units, WCS, products, and downstream use

SCI-MAP receives its scale and unit convention only through the realized PTC product; there is no direct SCI-CAL fallback. It preserves the coordinates, exact frame metadata, geometry, and coordinate validity supplied by ALIGN/AST through `SCI-AST:rtc_output_grid_coordinates@1` for the same stable RTC sample and parent chain. It records the complete declared frame identity— including the actual reference system, equinox/epoch, time scale, topology, WCS, axes, signs, and handedness—and does not replace it with "J2000." PTC owns transformed-sample identity and availability, producer-local validity, and MAP-facing coefficient/covariance facts. MAP owns the target grid and its own projection, coefficient-admission, boundary, support, and use conditions. VAL may represent and evaluate named rules; it does not author those policies.

The accepted map-bundle authority is ADR 0009 and its 2026-08-05 owner amendment. The full binary64 typed or sidecar WCS is the lossless identity used for admission, common-grid comparison, and provenance. The WCS serialized in the FITS product is the representation ordinary scientific tools are expected to use for pixel-to-sky conversion. It must preserve axis sign, handedness, orientation, and the exact centered-integer reference-pixel relations, and must agree with the lossless authority within 0.1 arcsec maximum sky separation. The bound is cited from that owner-approved representation authority; this rationale does not invent a physical justification for it.

The FITS WCS is sufficient for ordinary pixel-to-sky use at its declared serialization fidelity, while the sidecar retains the exact WCS identity used for admission, conformance, provenance, and coadd comparison. A tolerance never authorizes a different sky grid, fractional shift, interpolation, or reprojection.

Science-user state	Interpretation and required action
Valid map; response available	Use only for the response class and processing lineage actually declared.
Valid numerical map; response unavailable	Preserve the honest limitation. The original bundle makes no unsupported response-dependent claim; later analysis may attach a new versioned response or corrected product.
Formal weight; empirical significance unavailable	Use only according to its declared role—normalization or formal precision—and do not call it significance.
Covariance summarized or unavailable	State meaning, domain, and limits. The map remains valid; later covariance work is a new bound product.
Unsupported pixel	Unavailable, not zero sky.
MAP-invalid pixel	A downstream product may record separate use-local validity, but may not rewrite MAP-local base-product validity.
Coadd-incompatible observation	Entire observation bundle is excluded; no partial coadd.
WCS or unit incompatibility	Reject combination rather than silently convert, recenter, or relabel.

Downstream packages inherit the base/unfiltered MAP unit, response, covariance, validity, WCS, and provenance. NOI owns empirical noise and significance; FLT owns filtered response, covariance, and validity while preserving that parent; source, Pointing/OOF, and Beammap packages own their scientific inference.

8 Validation

Validation must keep algebra, software conformance, representation fidelity, observational performance, and production readiness separate. The required science program is:

1. analytic estimator tests for constants, unequal coefficients, and exact hand calculations;
2. exact one-hot half-open-boundary tests and explicit rejection of fractional projection until a successor family is authorized;
3. support, boundary, global-population, and unsupported-pixel tests;
4. complete-bundle compatibility and centered-integer coadd placement tests;
5. sequential/parallel conformance: floating results agree within a preregistered bound appropriate to operation count and conditioning, while identities, counts, masks, support, offsets, and product cardinality agree exactly;
6. synthetic point, extended, gradient, edge, and variable-coverage response tests; and
7. observational amplitude, shape, position, covariance, and repeatability checks against independent references.

No test, implementation inspection, reduction, or production qualification was executed for this authorship revision.

9 Open scientific decisions

The compact register in Appendix B preserves OD-001 through OD-009. SCI-MAP-CI-001 is resolved: `coverage_cut` is dimensionless. OD-007 remains open only for its numerical domain and failure behavior. OD-008 is resolved: the sole base projection is `SCI-MAP:one_hot_containing_pixel@1`, with lower-inclusive, upper-exclusive cells, outer-upper-boundary loss, no wrap or clamp, and fractional projection deferred. Eight MAP-local decisions remain open.

Rationale r0.6 completes the targeted scientific-closure pass. Further revisions should follow an owner decision, a normative contract change, new validation evidence, or a newly identified scientific inconsistency–not stylistic churn.

References

[1] *SCI-MAP v0.1 Formal Scientific/Engineering Contract*, canonical shared authority and r0.6 formal view.
[2] Frozen *SCI-PTC v0.1/r0.5* and the WP-7 scientific-owner dispositions bound by `SOURCE_MANIFEST_R0.6.md`.
[3] Current *SCI-CAL v0.1*, *SCI-AST v0.1/r0.3*, and *SCI-VAL v0.1/r0.3* Core/Registry/source bindings listed in that manifest.
[4] `SCI-PTC_TO_SCI-MAPv0.1/r0.1`; the source-current MAP upstream-admission and observation-coadd-admission profiles listed in the manifest.
[5] *Citlali Scientific Conventions* and ADR 0009, including its WCS/FITS persistence amendment.

A Compact rationale-to-contract crosswalk

The separate `CROSSWALK.md` carries the detailed routing. The formal contract remains normative; this appendix does not replace or paraphrase any requirement or prediction as authority.

Formal authority	Science-team location	Continuing formal home
REQ-001–010	Executive summary; Sections 1–2	Version, input, identity, projection, coefficient, lifecycle, admission.
REQ-011–018	Sections 2–3	Accumulators, valid-domain publication, response, absolute mode.
REQ-019–024	Section 4	Conditional covariance, formal weight, nuisance state, statistical labels.
REQ-025–035	Section 5	Separate incidence, contribution, exposure, support, and MAP-local validity facts.
REQ-036–042	Section 6	Complete bundles, centered-integer coadd, response/covariance, GLS boundary.
REQ-043–048	Section 7	Product identity, WCS, provenance, failure, publication.
REQ-049–052	Sections 7–8	Base MAP parent, consumer boundary, fixed realization operator, claim layers.
PRED-001–025	Section 8	Full formal prediction inventory and exact evidence classes.
OD-001–009	Section 9; Appendix B	External scientific-owner decision ledger.
SCI-MAP-CI-001	Section 5	Resolved dimensionless classification and bounded normative amendment.

B Compact scientific-owner decision register

Eight items are open and OD-008 is resolved. The exact questions, interim behavior, authority needed, and affected clauses live in `SCIENTIFIC_OWNER_DECISION_LEDGER.md`.

ID	Issue	Conservative status while open
OD-001	Threshold rationale	Adopted policy only; no physical optimality or completeness claim.
OD-002	Threshold change authority	Formula and population are frozen until an owner-approved successor.
OD-003	Response obligations	Original claims use only response information actually provided; later response work is a bound versioned product.
OD-004	Covariance persistence	Representation, meaning, domain, and omissions are explicit; absence does not invalidate the map.
OD-005	Observation-map publication	Follow the effective required-output policy; infer no universal storage mandate.
OD-006	Pointing/OOF reuse	No mode-specific authority follows from ordinary arithmetic alone.
OD-007	`coverage_cut` numeric domain	Dimensionless status is fixed; no numerical domain or boundary behavior is inferred.
OD-008	Projection normalization and boundaries	**RESOLVED**: one-hot containing-pixel only; half-open cells; outer-upper-boundary loss; no wrap/clamp; fractional projection deferred.
OD-009	Common-grid preparation and future mosaicking	Odd-difference inputs reject; MAP performs no crop, pad, reprojection, or mosaic absent new authority.

C Document authority

Rationale revision r0.6 integrates the targeted scientific closure and retains the scientific-owner disposition of SCI-MAP-CI-001. The scientific contract remains v0.1, with 52 canonical requirement IDs and 25 canonical prediction IDs unchanged. Their bounded amended clauses now classify `coverage_cut` as dimensionless while leaving its numerical domain and failure behavior open under OD-007. The formal contract and engineering conformance specification remain the normative authorities for audit, repair, re-audit, and exact product behavior.